

CMSC 471 Programming Assignment 1 Written Analysis**Question 1:**

For each heuristic, the method I follow is finding the letter positions where the current word differs from the goal word. $D(n)$ is the count of those positions. For example, "dog" and "cat" differ in all three positions so $D(\text{dog}) = 3$.

Steps: For Steps, my heuristic is $h(n) = D(n)$. Each action changes only one letter, so every position that has a letter different from the goal must be changed at least once. Each action costs 1 so a solution must cost at minimum $D(n)$. The heuristic may underestimate the cost if extra moves are needed to pass through legal words, but it will not overestimate it.

Scrabble: For Scrabble, my heuristic adds the Scrabble values of the goal letters at the positions that differ. For dog $\rightarrow$ cat, the estimate is $\text{value}(c) + \text{value}(a) + \text{value}(t) = 3 + 1 + 1 = 5$. The solution must insert each of those letters and will pay the cost of each value at least once. Any other letters inserted will add to the cost, so this estimate can't overestimate the cost.

Frequency: If the current word is the goal, my heuristic returns 0. Otherwise it returns $D(n) + \text{rarity}(\text{goal})$. At least $D(n)$ actions are needed and each costs at least 1 because rarity values are nonnegative. The path must also enter the goal word, so it must pay the rarity cost of the goal word. The starting word does not add a cost either. Since the heuristic only counts unavoidable costs, it cannot overestimate the cost.

Question 2:

Word Pair	Cost Measure	UCS Cost	UCS nodes expanded	A* Cost	A* nodes expanded	Reduction
WHY $\rightarrow$ ASK	Steps	5	773	5	16	97.93%
WHY $\rightarrow$ ASK	Scrabble	9	765	9	13	98.30%
WHY $\rightarrow$ ASK	Frequency	38.8554	808	38.8554	634	21.53%

BEAR→DUCK	Steps	4	691	4	6	99.13%
BEAR→DUCK	Scrabble	11	2981	11	6	99.80%
BEAR→DUCK	Frequency	38.2402	1084	38.2402	302	72.14%

- 1) UCS and A* returned the same cost in all six runs.
- 2) Scrabble had the largest percentage reduction for both word pairs.
- 3) Frequency was the least informative for both pairs, as shown by the smaller percentage reductions.
- 4) The steps heuristic estimates how many letter changes are still needed. The scrabble heuristic estimates the cost of inserting the missing goal letters. Both give A* useful information about the leftover work. The frequency heuristic also counts the needed changes but only adds the goal word's rarity. IT cannot predict which intermediate words a path will use or how expensive they are which results in A* not having as big of an improvement for frequency as it did for steps and scrabble.

Question 3:

For steps the optimal cost matches in both directions. This is because reversing any path gives a valid path with the same number of moves. Therefore, the cheapest path costs the same in either direction.

For Scrabble, the optimal cost does not match in both directions. This is because the costs of the letters in the starting word and in the goal word are not the same. An example is going from dog→cog inserts c and costs 3, but going from cog→dog inserts d and costs 2.

For Frequency, the optimal cost does not match in both directions. This is because the rarity of the words is different. For example, dog→cog costs $1 + \text{rarity}(\text{cog}) = 12.393166$ but cog→dog costs $1 + \text{rarity}(\text{dog}) = 6.819885$.

Question 4:

The model I used was Claude Sonnet 5 with the medium setting.

This is the prompt I gave it: In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is

stored in the dictionary. The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal. By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

Claude proposes finding $r_{\min}$, the smallest rarity among dictionary words with the same length as the goal. The heuristic is:

If n is the goal: $h_1(n) = 0$

Otherwise: $h_1(n) = (h_0(n) - 1)(1 + r_{\min}) + 1 + \text{rarity}(\text{goal})$

The final move has to enter the goal so its cost is $1 + \text{rarity}(\text{goal})$. at least $h_0(n) - 1$ other moves are also required and each costs at least $1 + r_{\min}$.

Verification: Claude’s heuristic is admissible. Any path from the current word to the goal needs at least $h_0(n)$ moves because each move changes only one letter position. The final move costs $1 + \text{rarity}(\text{goal})$. Every earlier move enters a word of the same length and costs at least $1 + r_{\min}$. Therefore, any path costs at least $(h_0(n) - 1)(1 + r_{\min}) + 1 + \text{rarity}(\text{goal})$, which is exactly $h_1(n)$. It cannot overestimate the minimum remaining cost.

The heuristic is also at least as large as h_0 . For a word that is not the goal, the difference between them is $(h_0(n) - 1) * r_{\min} + \text{rarity}(\text{goal})$. This cannot be negative because rarity values are nonnegative. Both goal words in my experiments have positive rarity, so h_1 is strictly larger for their non-goal states. Claude slightly overstated when equality is possible: if $h_0(n)$ is 1 and the goal’s rarity is 0, the two heuristics are equal even if $r_{\min}$ is positive. The formula is still admissible, so no repair was needed.

Heuristic testing results:

Word Pair	Heuristic	Solution Cost	Nodes Expanded
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WHY → ASK	h0	38.8554	755
WHY → ASK	Claude's h1	38.8554	634
BEAR → DUCK	h0	38.2402	720
BEAR → DUCK	Claude's h1	38.2402	257

Both heuristics return the same solution costs and Claude's heuristic expands fewer nodes in both experiments. This supports using Claude's heuristic for these problems.